

→ this is more FeCrAl, where is Ferrite-only

Q1] Ideally Cr & Al in the F/M steels should be at 9 & 12%, respectively.

First, Cr/Al is ~~also~~ important, too high of Cr/Al will result in many unfavorable changes in the microstructure.

Too little Cr will induce alpha phases that cause embrittlement. While too low Cr, will hinder the corrosion resistance. 85

F/M steels have carbides that stabilize the micro structure but can be affected by the concentrations of C, Cr, Ti & other precipitates.

Metallic precipitates (MX_n) & intermetallics depend on the concentrations as well, & eventually we had 9-Cr F/M steels & 12-Cr F/M steels.

Finally, ~~low~~ reduced activation F/M steels is achieved by reducing the elements easily activated in the reactor environment.

Q2] Ferritic steels swells less due to three main reasons:-

① - BCC structure allows for larger volume regarding the interstitial sites, this allows for more room for recombination of defects to occur.

② - Vacancies are much more mobile in ferritic steels (0.55 eV in F vs 1.4 eV in A), this also allows more room for recombination.

③ They introduce more sinks for point defects too, by having more carbide phases,

Q3] DBTT, is the temperature where the material will lose most of its ductility below it. But above it, the material is very ~~ductile~~ ductile as the defects (specially, the dislocation loops) become very mobile.

For F/M steels, the embrittlement can start due to many reasons. Depletion of Cr will induce alpha phases which in turn cause embrittlement. Another is the radiation induced segregation which drives many of the ~~other~~ precipitates & carbides to the ~~the~~ lath interface, making it lose its coherence.

FeCrAl alloys promise great radiation resistance & corrosion resistance that they can practically work for LWRs as a part of advanced reactor fuel & in next gen reactors.

They usually have 5-25 wt% Cr & 2-8 wt% Al. The corrosion resistance is due to the Al_2O_3 passive layer while the radiation resistance is due to the stable BCC ferritic structure stabilized by the addition of Cr & Al.

However, it has been found that low Cr & Al concentrations won't produce the Al_2O_3 layer, & high Cr & Al (Al > 5%) will produce it. A higher Cr but low Al concentrations will also produce Al_2O_3 passive layer. - too much Cr/Al? 7/8

25] Oxide particles mainly act as a very strong sink, especially to vacancies. They attract point defects & keep the microstructure stable & coherent.

4/6

- strengthening.

26] Ni alloys show great corrosion resistance & great high temperature creep resistance. They also can be great in molten salts due to Ni's great salt corrosion resistance. They have good radiation resistance. 8/8

However, He embrittlement can be a ~~huge~~ huge challenge due to Ni-n reactions that produce He that moves to boundaries. Also, Ni softens later on during operation, but the ductility keeps on decreasing causing a lot of embrittlement with a limited hardening. Ni alloys are also significantly more expensive.

27] U-Si has a much higher U density than U-Al, making it suitable for the RERTC program. This allows the reactor to gain high fission densities at the lower enrichments. However, U-Si fuels are not as compatible with the Al matrix & cladding as U-Al fuels. The stoichiometry in U-Si has major effects too, resulting in harder fabrication than that of U-Al.

When it comes to swelling, U_3Si will have a different response ~~than~~ than U_3Si_2 . U_3Si generally will undergo much higher swelling in the fuel matrix, due to the high ~~ratio of~~ volume increase by the fission products, since it has higher ratios of U in the matrix. But the Si enriched phase of U_3Si_2 is different as the extra Si atoms will help trap the gas bubbles & fission products & limit the swelling.

- alter amorphous structure/properties

11/10

28] U-Mo gamma phase is meta-stable. That's due to the microstructure evolution at higher temperatures that, over time, result in a phase change.

Irradiation, via atomic displacements, will hinder the phase-transformation, causing the U-Mo fuel to have a dominant gamma-phase

- mostly 6/7

29] This gap affects the homogeneity of U & Mo within the fuel matrix. A non-homogeneous fuel matrix results in a non-homogeneous fission profile & ~~fission~~ fission bubbles distribution. So, it's important to mitigate this.

- Kind of... 5/6

30] We have two major trends for the bubbles.

- Low Burnup: at low burnup, lower bubble mobility, & many sinks that act as a sponge to this bubbles, this will result in ~~less~~ smaller gas bubbles & lower swelling.
 - High Burnup: here we will have a very high density of fission products that won't let the bubbles to reach the sinks like at lower burnups. Eventually, bubbles will grow larger, driving the swelling dramatically higher. The rate here is much higher (quadratic).
- fission gas supersaturation, grain refinement, intergranular bubbles
- 4/10

Q11) U-Mo & Al interaction layer is important as this is where U-Mo-Al compounds will form. Bubbles will segregate at this layer resulting in a very high swelling that results in the failure of the fuel. $\frac{7}{7}$

To mitigate, U-Mo sheet is coated with a layer of Zr ($\sim 25 \mu\text{m}$ thick) to ~~separate~~ separate the fuel from the cladding & stopping the U-Mo-Al layers to form.
- only monolithic design also Si or coatings for dispersion

Q12) Al has a relatively low melting temperature ($\sim 600^\circ\text{C}$). At low research reactor temps, it's stable as it's usually at $\sim 0.3 T_m$. But in a LWR, it will have to be in a higher temperature ($\sim 0.5 T_m$) which will cause mechanical failure probably & increased swelling risks.
- why is Al good? $\frac{4}{6}$